C Programming Practice Questions

Questions, Code and Sample Output

1. Sum of Two Matrices

Question

Sum of Two Matrices

Code

```
#include <stdio.h>

int main()
{
    int a[10][10], b[10][10], sum[10][10];
    int rows, cols, i, j;

    printf("Enter the number of rows: ");
    scanf("%d", &rows);

    printf("Enter the number of columns: ");
    scanf("%d", &cols);

    printf("\nEnter elements of First Matrix:\n");
    for(i = 0; i < rows; i++)
        for(j = 0; j < cols; j++)
            scanf("%d", &a[i][j]);

    printf("\nEnter elements of Second Matrix:\n");
    for(i = 0; i < rows; i++)
        for(j = 0; j < cols; j++)
            scanf("%d", &b[i][j]);

    for(i = 0; i < rows; i++)
        for(j = 0; j < cols; j++)
            sum[i][j] = a[i][j] + b[i][j];

    printf("\nSum of the two matrices:\n");
    for(i = 0; i < rows; i++)
    {
        for(j = 0; j < cols; j++)
            printf("%d\t", sum[i][j]);
        printf("\n");
    }

    return 0;
}
```

Output

```
Enter the number of rows: 2
Enter the number of columns: 2
```

```
Enter elements of First Matrix:
1 2
3 4
Enter elements of Second Matrix:
5 6
7 8
Sum of the two matrices:
6 8
10 12
```

2. Read n Values in an Array and Display in Reverse Order

Question

Read n Values in an Array and Display in Reverse Order

Code

```
#include <stdio.h>

int main()
{
    int arr[100], n, i;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    printf("Array elements in reverse order:\n");
    for(i = n - 1; i >= 0; i--)
        printf("%d ", arr[i]);

    printf("\n");
    return 0;
}
```

Output

```
Enter the number of elements: 5
Enter 5 elements:
10 20 30 40 50
Array elements in reverse order:
50 40 30 20 10
```

3. Count the Total Number of Duplicate Elements in an Array

Question

Count the Total Number of Duplicate Elements in an Array

Code

```
#include <stdio.h>

int main()
{
    int arr[100], n, i, j, count = 0;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    for(i = 0; i < n; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(arr[i] == arr[j])
            {
                count++;
                break;
            }
        }
    }

    printf("Total number of duplicate elements = %d\n", count);
    return 0;
}
```

Output

```
Enter the number of elements: 7
Enter 7 elements:
1 2 3 2 4 5 1
Total number of duplicate elements = 2
```

4. Print Unique Elements in an Array

Question

Print Unique Elements in an Array

Code

```
#include <stdio.h>

int main()
{
    int arr[100], n, i, j, count;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    printf("Unique elements in the array are:\n");

    for(i = 0; i < n; i++)
    {
        count = 0;
        for(j = 0; j < n; j++)
        {
            if(arr[i] == arr[j])
                count++;
        }

        if(count == 1)
            printf("%d ", arr[i]);
    }

    printf("\n");
    return 0;
}
```

Output

```
Enter the number of elements: 8
Enter 8 elements:
1 2 3 2 4 5 1 6
Unique elements in the array are:
3 4 5 6
```

5. Insert an Element in a Sorted Array

Question

Insert an Element in a Sorted Array

Code

```
#include <stdio.h>

int main()
{
    int arr[100], n, element, pos, i;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements in ascending order:\n", n);
    for(i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    printf("Enter the element to insert: ");
    scanf("%d", &element);

    pos = n;
    for(i = 0; i < n; i++)
    {
        if(arr[i] > element)
        {
            pos = i;
            break;
        }
    }

    for(i = n; i > pos; i--)
        arr[i] = arr[i - 1];

    arr[pos] = element;
    n++;

    printf("Array after insertion:\n");
    for(i = 0; i < n; i++)
        printf("%d ", arr[i]);

    printf("\n");
    return 0;
}
```

Output

```
Enter the number of elements: 5
Enter 5 elements in ascending order:
10 20 30 40 50
Enter the element to insert: 35
Array after insertion:
10 20 30 35 40 50
```

6. Insert an Element in an Unsorted Array

Question

Insert an Element in an Unsorted Array

Code

```
#include <stdio.h>

int main()
{
    int arr[100], n, element, position, i;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    printf("Enter the element to insert: ");
    scanf("%d", &element);

    printf("Enter the position (1 to %d): ", n + 1);
    scanf("%d", &position);

    if(position < 1 || position > n + 1)
    {
        printf("Invalid position!\n");
    }
    else
    {
        for(i = n; i >= position; i--)
            arr[i] = arr[i - 1];

        arr[position - 1] = element;
        n++;

        printf("Array after insertion:\n");
        for(i = 0; i < n; i++)
            printf("%d ", arr[i]);
        printf("\n");
    }

    return 0;
}
```

Output

```
Enter the number of elements: 5
Enter 5 elements:
10 30 20 50 40
Enter the element to insert: 25
Enter the position (1 to 6): 3
```

```
Array after insertion:
10 30 25 20 50 40
```

7. Matrix Multiplication

Question

Matrix Multiplication

Code

```
#include <stdio.h>

int main()
{
    int a[10][10], b[10][10], c[10][10];
    int r1, c1, r2, c2, i, j, k;

    printf("Enter rows and columns of First Matrix: ");
    scanf("%d%d", &r1, &c1);

    printf("Enter rows and columns of Second Matrix: ");
    scanf("%d%d", &r2, &c2);

    if(c1 != r2)
    {
        printf("Matrix multiplication is not possible.\n");
        return 0;
    }

    printf("Enter elements of First Matrix:\n");
    for(i = 0; i < r1; i++)
        for(j = 0; j < c1; j++)
            scanf("%d", &a[i][j]);

    printf("Enter elements of Second Matrix:\n");
    for(i = 0; i < r2; i++)
        for(j = 0; j < c2; j++)
            scanf("%d", &b[i][j]);

    for(i = 0; i < r1; i++)
        for(j = 0; j < c2; j++)
        {
            c[i][j] = 0;
            for(k = 0; k < c1; k++)
                c[i][j] += a[i][k] * b[k][j];
        }

    printf("\nProduct of the two matrices:\n");
    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c2; j++)
            printf("%d\t", c[i][j]);
        printf("\n");
    }

    return 0;
}
```

Output

```
Enter rows and columns of First Matrix: 2 2
Enter rows and columns of Second Matrix: 2 2
Enter elements of First Matrix:
1 2
3 4
Enter elements of Second Matrix:
5 6
7 8
Product of the two matrices:
19 22
43 50
```

8. Matrix Transpose

Question

Matrix Transpose

Code

```
#include <stdio.h>

int main()
{
    int a[10][10], transpose[10][10];
    int rows, cols, i, j;

    printf("Enter the number of rows: ");
    scanf("%d", &rows);

    printf("Enter the number of columns: ");
    scanf("%d", &cols);

    printf("Enter the elements of the matrix:\n");
    for(i = 0; i < rows; i++)
        for(j = 0; j < cols; j++)
            scanf("%d", &a[i][j]);

    for(i = 0; i < rows; i++)
        for(j = 0; j < cols; j++)
            transpose[j][i] = a[i][j];

    printf("\nTranspose of the matrix:\n");
    for(i = 0; i < cols; i++)
    {
        for(j = 0; j < rows; j++)
            printf("%d\t", transpose[i][j]);
        printf("\n");
    }

    return 0;
}
```

Output

```
Enter the number of rows: 2
Enter the number of columns: 3
Enter the elements of the matrix:
1 2 3
4 5 6
Transpose of the matrix:
1 4
2 5
3 6
```

9. Sum of the Right Diagonal of a Matrix

Question

Sum of the Right Diagonal of a Matrix

Code

```
#include <stdio.h>

int main()
{
    int a[10][10], n, i, j, sum = 0;

    printf("Enter the order of the square matrix: ");
    scanf("%d", &n);

    printf("Enter the elements of the matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < n; j++)
            scanf("%d", &a[i][j]);

    for(i = 0; i < n; i++)
        for(j = 0; j < n; j++)
            if(i + j == n - 1)
                sum += a[i][j];

    printf("Sum of the right diagonal = %d\n", sum);
    return 0;
}
```

Output

```
Enter the order of the square matrix: 3
Enter the elements of the matrix:
1 2 3
4 5 6
7 8 9
Sum of the right diagonal = 15
```

10. Check for Identity Matrix

Question

Check for Identity Matrix

Code

```
#include <stdio.h>

int main()
{
    int a[10][10], n, i, j, flag = 1;

    printf("Enter the order of the square matrix: ");
    scanf("%d", &n);

    printf("Enter the elements of the matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < n; j++)
            scanf("%d", &a[i][j]);

    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            if(i == j)
            {
                if(a[i][j] != 1)
                {
                    flag = 0;
                    break;
                }
            }
            else
            {
                if(a[i][j] != 0)
                {
                    flag = 0;
                    break;
                }
            }
        }

        if(flag == 0)
            break;
    }

    if(flag == 1)
        printf("The matrix is an Identity Matrix.\n");
    else
        printf("The matrix is NOT an Identity Matrix.\n");

    return 0;
}
```

Output

```
Enter the order of the square matrix: 3
Enter the elements of the matrix:
1 0 0
0 1 0
0 0 1
The matrix is an Identity Matrix.
```

11. Count Vowels, Consonants, Digits and White Spaces

Question

Count Vowels, Consonants, Digits and White Spaces

Code

```
#include <stdio.h>

int main()
{
    char str[100];
    int i;
    int vowels = 0, consonants = 0, digits = 0, spaces = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if((str[i] >= 'A' && str[i] <= 'Z') ||
           (str[i] >= 'a' && str[i] <= 'z'))
        {
            if(str[i] == 'A' || str[i] == 'E' || str[i] == 'I' ||
               str[i] == 'O' || str[i] == 'U' ||
               str[i] == 'a' || str[i] == 'e' || str[i] == 'i' ||
               str[i] == 'o' || str[i] == 'u')
                vowels++;
            else
                consonants++;
        }
        else if(str[i] >= '0' && str[i] <= '9')
            digits++;
        else if(str[i] == ' ')
            spaces++;
    }

    printf("Number of vowels       = %d\n", vowels);
    printf("Number of consonants   = %d\n", consonants);
    printf("Number of digits       = %d\n", digits);
    printf("Number of white spaces = %d\n", spaces);

    return 0;
}
```

Output

```
Enter a string: Hello World 123
Number of vowels       = 3
Number of consonants   = 7
Number of digits       = 3
Number of white spaces = 2
```

12. Find the Frequency of a Character in a String

Question

Find the Frequency of a Character in a String

Code

```
#include <stdio.h>

int main()
{
    char str[100], ch;
    int i, count = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    printf("Enter the character to find: ");
    scanf("%c", &ch);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] == ch)
            count++;
    }

    printf("Frequency of '%c' = %d\n", ch, count);
    return 0;
}
```

Output

```
Enter a string: programming
Enter the character to find: g
Frequency of 'g' = 2
```

13. Count the Words in a String

Question

Count the Words in a String

Code

```
#include <stdio.h>

int main()
{
    char str[100];
    int i, words = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if((i == 0 && str[i] != ' ' && str[i] != '\n') ||
           (str[i] != ' ' && str[i - 1] == ' '))
            words++;
    }

    printf("Number of words = %d\n", words);
    return 0;
}
```

Output

```
Enter a string: C programming is easy
Number of words = 4
```

14. Sort an Array of Strings in Alphabetical Order

Question

Sort an Array of Strings in Alphabetical Order

Code

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[10][100], temp[100];
    int n, i, j;

    printf("Enter the number of strings: ");
    scanf("%d", &n);
    getchar();

    printf("Enter %d strings:\n", n);
    for(i = 0; i < n; i++)
    {
        fgets(str[i], sizeof(str[i]), stdin);
        str[i][strcspn(str[i], "\n")] = '\0';
    }

    for(i = 0; i < n - 1; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(strcmp(str[i], str[j]) > 0)
            {
                strcpy(temp, str[i]);
                strcpy(str[i], str[j]);
                strcpy(str[j], temp);
            }
        }
    }

    printf("\nStrings in alphabetical order:\n");
    for(i = 0; i < n; i++)
        printf("%s\n", str[i]);

    return 0;
}
```

Output

```
Enter the number of strings: 5
Enter 5 strings:
Banana
Apple
Orange
Mango
Grapes
```

```
Strings in alphabetical order:
Apple
Banana
Grapes
Mango
Orange
```

15. Toggle Case of a Sentence

Question

Toggle Case of a Sentence

Code

```
#include <stdio.h>

int main()
{
    char str[100];
    int i;

    printf("Enter a sentence: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] >= 'A' && str[i] <= 'Z')
            str[i] = str[i] + 32;
        else if(str[i] >= 'a' && str[i] <= 'z')
            str[i] = str[i] - 32;
    }

    printf("Sentence after toggling case:\n%s", str);
    return 0;
}
```

Output

```
Enter a sentence: Hello World 123
Sentence after toggling case:
hELLO wORLD 123
```